

The Job Sequencing and Tool Switching Problem

Part IV: GTSP-MTZ Formulation for the Non-Compact SSP

Replacing Exponential Subtour Constraints with Polynomial MTZ Constraints

Motivation: Solving the Double-Generation Problem

The non-compact ILP of Part II (`02_noncompact_formulation.tex`) has two simultaneously exponential components:

- **Exponential columns:** $|\mathcal{C}| = \binom{m}{b}$ configuration variables $\rightarrow$ requires **column generation** (H-SUKP pricing).
- **Exponential rows:** PCTSP-based subtour elimination constraints $\rightarrow$ requires **row generation** (separation via cut-finding).

Running both simultaneously (“double generation”) is complex: adding a new column can re-violate previously satisfied subtour cuts, forcing repeated separation passes. This interaction makes the LP solve at each Branch-and-Price node very expensive.

This document’s solution: replace the exponential subtour constraints with a polynomial ($O(n^2)$) set of *Miller–Tucker–Zemlin* (MTZ) constraints adapted to the cluster structure of the GTSP. Row generation is completely eliminated; only column generation remains. The trade-off is:

	Old (SEC-based)	New (MTZ-based)
Variables	$ V ^2$ (exponential)	$ V ^2$ (exponential)
Subtour constraints	$2^{ V }$ (exponential)	n^2 (polynomial)
Coverage constraints	n	$2n$
Column generation needed?	Yes	Yes
Row generation needed?	Yes	No

Row generation is eliminated. Column generation for the exponential variable set $\{x_{ij}\}$ remains necessary.

Notation

- $J = \{1, \dots, n\}$: set of n jobs; T : set of tools; C : magazine capacity.
- $T_j \subseteq T$, $|T_j| \leq C$: mandatory tools of job j .
- V : set of **all valid configurations** — all subsets $\mathcal{C} \subseteq T$ with $|\mathcal{C}| = C$. $|V| = \binom{|T|}{C}$, exponential in general.
- For each job $j^* \in J$, define its **configuration cluster**:

$$H_{j^*} = \{\mathcal{C} \in V : T_{j^*} \subseteq \mathcal{C}\},$$

the set of all configurations in V that can serve job j^* .

- **Switching cost** between configurations $i, j \in V$:

$$d_{ij} = C - |i \cap j| = |i \setminus j|.$$

- $n = |J|$: number of jobs (used in MTZ coefficient).
- **Dummy nodes** $s, t \notin V$: artificial source and sink for the path formulation (connected to all $i \in V$ with zero-cost arcs).

The Complete GTSP-MTZ Formulation

Let $V^+ = V \cup \{s, t\}$ (extended node set). Define $x_{ij} \in \{0, 1\}$ for all $(i, j) \in V^+ \times V^+, i \neq j$.

GTSP-MTZ Formulation for SSP (Path version)

Objective:

$$\min \sum_{i \in V} \sum_{j \in V, j \neq i} d_{ij} x_{ij} \quad (1)$$

Path start and end (dummy source s and sink t):

$$\sum_{j \in V} x_{sj} = 1 \quad (2)$$

$$\sum_{i \in V} x_{it} = 1 \quad (3)$$

Configuration-level degree constraints (each config used at most once):

$$\sum_{j \in V^+ \setminus \{i\}} x_{ij} \leq 1 \quad \forall i \in V \quad (4)$$

$$\sum_{i \in V^+ \setminus \{j\}} x_{ij} \leq 1 \quad \forall j \in V \quad (5)$$

Flow conservation (if a config is entered, it is also exited):

$$\sum_{j \in V^+ \setminus \{i\}} x_{ij} = \sum_{j \in V^+ \setminus \{i\}} x_{ji} \quad \forall i \in V \quad (6)$$

Cluster coverage — Exit constraint (the path must leave each job's cluster):

$$\sum_{i \in H_{j^*}} \sum_{j \in V \setminus H_{j^*}} x_{ij} \geq 1 \quad \forall j^* \in J \quad (7)$$

Cluster coverage — Entry constraint (the path must enter each job's cluster):

$$\sum_{i \in V \setminus H_{j^*}} \sum_{j \in H_{j^*}} x_{ij} \geq 1 \quad \forall j^* \in J \quad (8)$$

Cluster-level MTZ subtour elimination:

$$u_{j'} - u_{j''} + n \sum_{i \in H_{j'}} \sum_{j \in H_{j''}} x_{ij} \leq n - 1 \quad \forall j', j'' \in J, j' \neq j'' \quad (9)$$

Bounds on position variables:

$$1 \leq u_{j'} \leq n \quad \forall j' \in J \quad (10)$$

Integrality:

$$x_{ij} \in \{0, 1\} \quad \forall i, j \in V^+, i \neq j \quad (11)$$

Mathematical Verification**Correctness of the Objective**

The objective (1) sums $d_{ij} = C - |i \cap j|$ over all arcs $(i \rightarrow j)$ selected in the solution. Since $|i| = |j| = C$, we have $|i \setminus j| = C - |i \cap j|$ (tools removed when transitioning from i to j) = tools inserted (uniform SSP model). The objective correctly minimises total tool switches.

Correctness of Coverage Constraints

Proposition 4.1 (Coverage $\Leftrightarrow$ Job Feasibility). *In any feasible solution to the formulation, every job $j^* \in J$ is served: there exists at least one visited configuration $\mathcal{C}$ with $T_{j^*} \subseteq \mathcal{C}$ (i.e., $\mathcal{C} \in H_{j^*}$).*

Proof. The selected arcs $\{(i, j) : x_{ij} = 1\}$ form a path from s to t through a sequence of configurations $\mathcal{C}_{(1)}, \mathcal{C}_{(2)}, \dots$. The exit constraint (7) requires at least one arc from inside H_{j^*} to outside H_{j^*} , and the entry constraint (8) requires at least one arc from outside H_{j^*} into H_{j^*} . Since the selected arcs form a connected path, the existence of both an entry and an exit arc for H_{j^*} implies the path passes through at least one node in H_{j^*} , i.e., at least one visited configuration belongs to H_{j^*} . That configuration serves job j^* . $\square$ $\square$

Correctness of MTZ Subtour Elimination

Proposition 4.2 (MTZ Eliminates Cluster-Level Subtours). *In any feasible integer solution, the selected arcs do not contain a closed tour visiting only a strict subset of job clusters. More precisely: if there exists a directed cycle in the arc solution that visits configurations only from clusters $\{H_{j'} : j' \in S\}$ for some $S \subsetneq J$, then constraint (9) is violated.*

Proof. Suppose for contradiction there is a directed cycle among configurations visiting clusters $S \subsetneq J$. Then for each consecutive cluster pair (j', j'') in this cycle, there exists an arc (i, j) with $i \in H_{j'}$, $j \in H_{j''}$ with $x_{ij} = 1$, so $\sum_{i \in H_{j'}} \sum_{j \in H_{j''}} x_{ij} \geq 1$.

Summing constraint (9) around the cycle for all consecutive pairs (j', j'') in S :

$$\sum_{\text{cycle}} (u_{j'} - u_{j''}) + n \sum_{\text{cycle}} \sum_{i \in H_{j'}} \sum_{j \in H_{j''}} x_{ij} \leq |S|(n - 1).$$

The left-hand telescoping sum of u -differences around a cycle equals zero. The flow term gives $\sum_{\text{cycle}} 1 = |S|$ (at least one arc per cluster pair). Hence $n|S| \leq |S|(n - 1)$, i.e. $n \leq n - 1$, a contradiction. Therefore no cluster-level subtour exists. $\square$ $\square$

Important caveat — Intra-cluster subtours. Constraint (9) eliminates subtours at the *cluster level* but does **not** eliminate subtours involving multiple configurations from the *same* cluster H_{j^*} (i.e. a cycle entirely within H_{j^*}).

However, such intra-cluster subtours are **never optimal** for SSP: visiting two configurations $\mathcal{C}_a, \mathcal{C}_b \in H_{j^*}$ in a cycle serves job j^* redundantly (twice) and incurs an avoidable switching cost. Any optimal integer solution will not exhibit intra-cluster cycles. The LP relaxation may, but the integer optimal will not.

If a stronger guarantee is needed (e.g. for LP bound quality), add configuration-level MTZ constraints:

$$v_i - v_j + Nx_{ij} \leq N - 1 \quad \forall i, j \in V$$

with $N = |V|$ and $v_i \in \mathbb{R}$. These eliminate all subtours but reintroduce $O(|V|^2)$ (exponential) constraints.

Correctness as Path (not Cycle)

Proposition 4.3 (Path Structure). *The dummy-node constraints (2)–(3) together with (6) ensure the selected arcs form a directed path from s to t through some subset of V .*

Proof. Constraint (2): exactly one arc leaves s . Constraint (3): exactly one arc enters t . Constraint (6): for each internal node $i \in V$, in-degree = out-degree. The degree constraints (4)–(5) bound in- and out-degree to ≤ 1 . Together, these conditions define a collection of directed paths and cycles from s to t . The MTZ constraints (9) eliminate cycles among the clusters J . Since s and t are distinct dummy nodes not in any cluster H_{j^*} , the solution is a single directed path $s \rightarrow \mathcal{C}_{(1)} \rightarrow \dots \rightarrow \mathcal{C}_{(k)} \rightarrow t$. $\square$ $\square$

Equivalence to OPT-SSP

Theorem 4.4 (Formulation Correctness). *The optimal objective value of the GTSP-MTZ formulation (1)–(11) equals OPT_{SSP} : the minimum total tool switches over all feasible job orderings and all valid configuration sequences.*

Proof. (Feasibility $\Rightarrow$ SSP solution.) Any integer feasible solution gives a path $s \rightarrow \mathcal{C}_{(1)} \rightarrow \dots \rightarrow \mathcal{C}_{(k)} \rightarrow t$ through configurations. By proposition 4.1, every job $j^* \in J$ has $\mathcal{C}_{(l)} \in H_{j^*}$ for some l ; process job j^* when configuration $\mathcal{C}_{(l)}$ is active. The total switching cost is $\sum_{l=1}^{k-1} d_{\mathcal{C}_{(l)}, \mathcal{C}_{(l+1)}}$ = objective value. This is a feasible SSP solution of the same cost.

(SSP solution $\Rightarrow$ Feasibility.) Any feasible SSP solution processes jobs in some order with configurations $\mathcal{C}^{(1)}, \dots, \mathcal{C}^{(n)}$ (one per job, possibly repeated). Collapsing consecutive identical configurations, we get a sequence $\mathcal{C}_{(1)}, \dots, \mathcal{C}_{(k)}$ of distinct consecutive configurations. Set $x_{\mathcal{C}_{(l)}, \mathcal{C}_{(l+1)}} = 1$ for $l = 1, \dots, k-1$, and $x_{s, \mathcal{C}_{(1)}} = x_{\mathcal{C}_{(k)}, t} = 1$. All constraints (2)–(11) are satisfied: degree constraints hold by the path structure; coverage holds since every job is served; MTZ is satisfied by setting $u_{j^*} = \min\{l : \mathcal{C}_{(l)} \in H_{j^*}\}$. The objective equals the SSP solution cost.

Optimality follows because the bijection above preserves costs. $\square$ $\square$

Constraint Count and Complexity

Constraint group	Description	Count
(2)–(3)	Path start/end (dummy nodes)	2
(4)–(5)	Config-level degree (at most one in/out)	$2 V $
(6)	Flow conservation at each config	$ V $
(7)	Exit coverage per job cluster	n
(8)	Entry coverage per job cluster	n
(9)	Cluster-level MTZ subtour elimination	$n(n-1)$
(10)	Bounds on $u_{j'}$	$2n$
Total		$3 V + n(n+1) + 4$

The constraints in $|V|$ are unavoidable (one per variable column) and handled by column generation. The constraints in n — the MTZ and coverage groups — are **polynomial in the number of jobs**: $O(n^2)$. This is the key improvement over the SEC-based formulation, which has $O(2^{|V|})$ subtour constraints requiring row generation.

Remark 5.1 (Column Generation). The decision variables x_{ij} are indexed by pairs of configurations. Since $|V| = \binom{T}{C}$ is exponential, column generation is still necessary. The pricing problem: given dual variables (μ_i, λ_j) for the degree constraints and (π_{j^*}) for the coverage constraints, find the arc (i, j) with most negative reduced cost $\bar{d}_{ij} = d_{ij} - \mu_i - \lambda_j - \sum_{j^*: i \in H_{j^*}, j \notin H_{j^*}} \pi_{j^*}$. This reduces to finding, for each pair of job clusters, the pair of configurations $(i \in H_{j'}, j \in H_{j''})$ minimising switching cost d_{ij} minus dual bonuses. For structured instances (e.g. laminar tool families), this is tractable.

Comparison with the Board's Formulation

The board's formulation is mathematically correct in its core ideas. The complete formulation above adds three elements that were implicit or missing:

- (1) **Dummy source/sink nodes** (2)–(3): The board uses the cycle-TSP structure (degree = 1 for all nodes). The path version (SSP is a path, not a cycle) requires dummy nodes. Without them, the formulation finds a Hamiltonian *cycle* which is not what SSP requires. A valid alternative: set $d_{ts} = 0$ (free return arc) and minimise the cycle cost, then break at $(t \rightarrow s)$.
- (2) **Flow conservation** (6): Ensures that if a configuration is entered along the path, it is also exited (the path does not terminate mid-way at an internal node).
- (3) **Configuration-level degree constraints** (4)–(5): Without these, multiple arcs could leave or enter the same configuration, which would not correspond to a valid sequential job processing order.

The entry/exit constraints (7)–(8) and the cluster-level MTZ (9) are exactly as on the board and are correct.

Remark 6.1 (Why n in the MTZ coefficient). The MTZ constraint uses $n = |J|$ (number of jobs/clusters) rather than $|V|$ (number of configurations). This is correct because the position variables $u_{j'}$ track the *order of cluster visits*, not configurations. There are at most n clusters to visit, so $u_{j'} \in [1, n]$ suffices. Using $|V|$ instead would be a valid but unnecessarily

weak (larger) coefficient, relaxing the constraint and potentially worsening the LP bound.

Summary

The GTSP-MTZ formulation for SSP:

- Is **mathematically correct**: theorem 4.4 proves it exactly captures OPT_{SSP} .
- Has **polynomial subtour constraints**: $O(n^2)$ MTZ constraints replace $O(2^{|V|})$ SEC constraints — row generation is eliminated.
- Still requires **column generation**: the variable set $\{x_{ij}\}$ is exponential in size.
- Has a **known LP bound weakness**: cluster-level MTZ is weaker than SEC for the LP relaxation (larger integrality gap), as is standard for MTZ vs. SEC in classic TSP.
- Can be **strengthened** by: adding the JGP-grouping structure (restricting V to JGP-feasible configs), using Lifted MTZ constraints, or incorporating a MWHP-based ordering warm start.

Where This Fits: Solution Architecture

The four formulation families form a progression in tractability:

Formulation	Column gen.?	Row gen.?
Compact ILPs (Part I)	No	No
Non-compact + PCTSP (Part II)	Yes	Yes
MTZ (this document)	Yes	No
Branch-and-Benders	No	Yes (lazy cuts)

The MTZ formulation eliminates row generation entirely: with only column generation, a Branch-and-Price approach becomes much more manageable. The trade-off is a weaker LP bound vs. PCTSP-based subtour elimination—the same MTZ vs. SEC trade-off as in the classical TSP [Applegate et al.(2006)Applegate, Bixby, Chvátal, and Cook].

References

[Applegate et al.(2006)Applegate, Bixby, Chvátal, and Cook] D. L. Applegate, R. E. Bixby, V. Chvátal, and W. J. Cook. *The Traveling Salesman Problem: A Computational Study*. Princeton University Press, 2006.